A Neural Shrimp and Its Activity

Numerical Reproduction in a Three-Neuron Recurrent Network

Authorship to be finalized by the human researcher

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Private exploratory working draft. EXP-004–005 reproduce and probe a published small recurrent network. Connectivity is fixed during each trajectory; no training, reasoning benchmark, or task-accuracy benefit is claimed.

Abstract

We numerically reproduce a shrimp-shaped periodic window in the connection-weight plane of the three-neuron tanh network studied by da Silva and Rech. A frozen experiment measures 39,322 grid settings and a separate 201-point transect. Independent shooting verifies two attracting nine-peak orbits and a nearby eighteen-peak orbit. At the core, the complete state and activation waveform repeat every 37.63637538 time units; the dominant transverse Floquet multiplier is 0.06506, alongside the required neutral flow mode. A separately frozen follow-up uses independent initialization and 10,000 units of settling at two integration steps. Its two distant controls retain absent short recurrence and positive finite-time exponents. After a 10^{-6} state perturbation and 300 time units, activation-vector separation is 1.53×10^{-6} at the core versus 1.41 and 2.13 at those controls. Thus the geometry accompanies a large, directly measured difference in neural waveform repeatability. The original near-edge exterior fails its intended interpretation and is retained: refinement makes it periodic, and the follow-up independently verifies that orbit. These are finite-horizon numerical observations in a fixed-weight recurrent network. The next utility test is whether window geometry predicts tolerance to connection-weight drift beyond local cycle diagnostics; task benefit has not yet been measured.

1 The measured object

Figure 1 is a map of the neural network itself. Its controls are two connection weights, and its colors describe recurring sequences of the complete three-neuron state. The focused window has a broad nine-peak body and narrow extensions; selected nine- and eighteen-peak trajectories pass independent orbit checks. This is a numerical reproduction of the small-network precedent of da Silva and Rech [1], not a first report of neural shrimps. The term refers to periodic-window organization in a control plane [2]; the corresponding trajectories evolve in state space.

Our question is what this geometry means for the parent network. We therefore place recurrence, sensitivity, neuron potentials, and actual tanh activations side by side. Modern training maps and latent-reasoning basin maps motivate the broader question [3, 4], but the present result concerns the fixed-weight continuous-time system below.

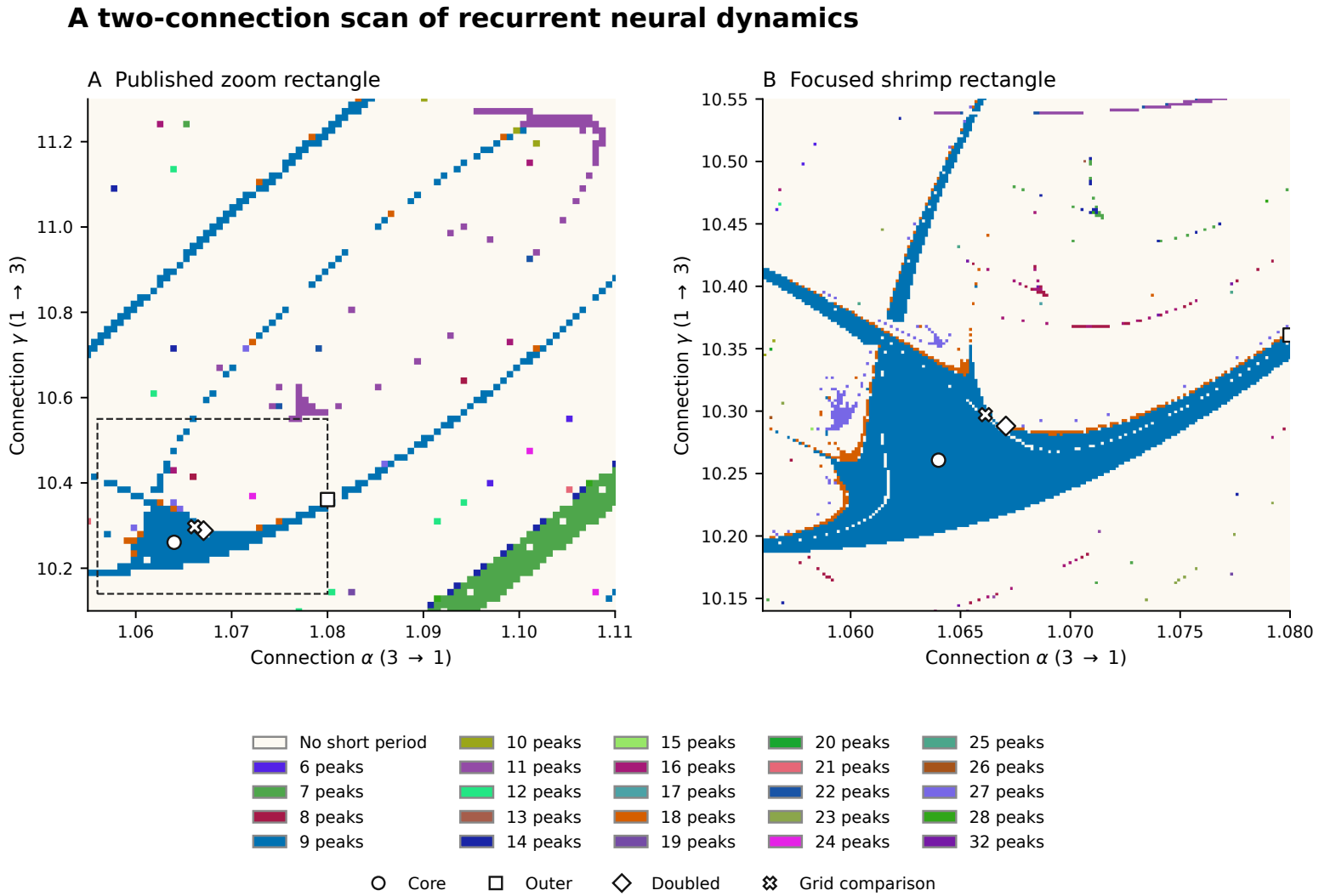


Figure 1: Measured neural parameter map. (A) The published zoom rectangle sampled on an 81×81 grid. (B) The predetermined 181×181 focus resolves the nine-peak shrimp body and extensions, with other periodic candidates nearby. Every cell comes from the neural ODE; there is no drawn shrimp or interpolated reconstruction. Colors count peaks in a recurring *full-state* event sequence, rather than time steps. Pale cells have no detected short period, not a chaos certificate. Markers locate the frozen core, outer-component, doubled, and grid-comparison representatives (the last was originally named exterior). The outer point lies on the focus boundary; its label does not certify a limb tip. The exterior label records coarse-grid selection and fails its intended interpretation under refinement (Table 1).

2 Network, scan, and orbit verification

2.1 The parent network and its readout

The model is the asymmetric three-neuron system in da Silva and Rech [1]:

$$\dot{x} = -x + W\tanh(x), \qquad W = \begin{pmatrix} 1.5 & 2.9 & \alpha \\ -1.5 & 1.18 & 0 \\ \gamma & -22 & 0.47 \end{pmatrix}, \qquad x \in \mathbb{R}^3. \tag{1}$$

The coordinates x_i are neuron potentials, and $u_i = \tanh(x_i)$ are their activation outputs. The weights α and γ respectively couple neurons $3 \to 1$ and $1 \to 3$. They are constant during a run. Thus the experiment changes network connectivity between settings, not a learning rate while training.

2.2 A frozen exploratory reproduction

The overview spans $\alpha \in [1.055, 1.110]$ and $\gamma \in [10.1, 11.3]$ at 81^2 settings. The focus spans $[1.056, 1.080] \times [10.14, 10.55]$ at 181^2 settings. Classical fourth-order Runge–Kutta uses $\Delta t = 0.02$, a settling time of 2,000, and a measurement time of 800. Each increasing-α row begins at $x = (0.1, 0.1, 0.1)$ and then continues from the previous setting's terminal state. Rows are independent. The focused rectangle was selected from the published figure before outcomes existed.

A separate transect fixes $\alpha = 1.064$ and samples 201 values of $\gamma \in [10.14, 10.55]$. Each point independently starts from the same prescribed state, using $\Delta t = 0.01$, settling time 4,000, and measurement time 1,000. It therefore checks whether the main observed structure survives a common initialization without inheriting grid continuation.

A maximum of x_3 defines an event, localized by cubic Hermite interpolation of the numerical endpoints. The retained event includes time and all three state coordinates. The screen searches the last 128 events for four repetitions of a primitive sequence of at most 32 peaks, with absolute tolerance 2×10^{-6} and relative tolerance 2×10^{-5}. A zero label means no short period passed this screen. Finite-time Lyapunov estimates evolve one tangent vector, discard 100 units of alignment, and normalize growth by the remaining measurement time. They are finite-horizon measurements; a stable autonomous cycle has a neutral tangent direction.

2.3 Verification of selected trajectories

The frozen rules select the period-nine candidate nearest $(1.064, 10.26)$, the farthest member of its eight-connected period-nine component, the nearest period-eighteen candidate, and a nearby no-short-period cell with positive finite-time exponent. All four are re-integrated from their captured terminal states at $\Delta t = 0.01$ and 0.005, each with settling time 4,000 and measurement time 1,200. These refinements test the captured regime, not basin independence. Original labels and changed classifications are both retained.

Independent adaptive DOP853 shooting solves $\phi_T(x_0) = x_0$ with an x_3-maximum phase condition, relative state closure at most 10^{-8}, and the expected primitive peak count. Variational integration gives $M(T) = D\phi_T(x_0)$. One multiplier must be neutral and aligned with the flow; both transverse multipliers must have modulus below $1 - 10^{-5}$. Neutral error and flow residual must each be at most 10^{-6}, with alignment at least 0.99. Solver tolerances are 10^{-10} relative and 10^{-12} absolute. These are numerical orbit certificates, not interval-arithmetic proofs.

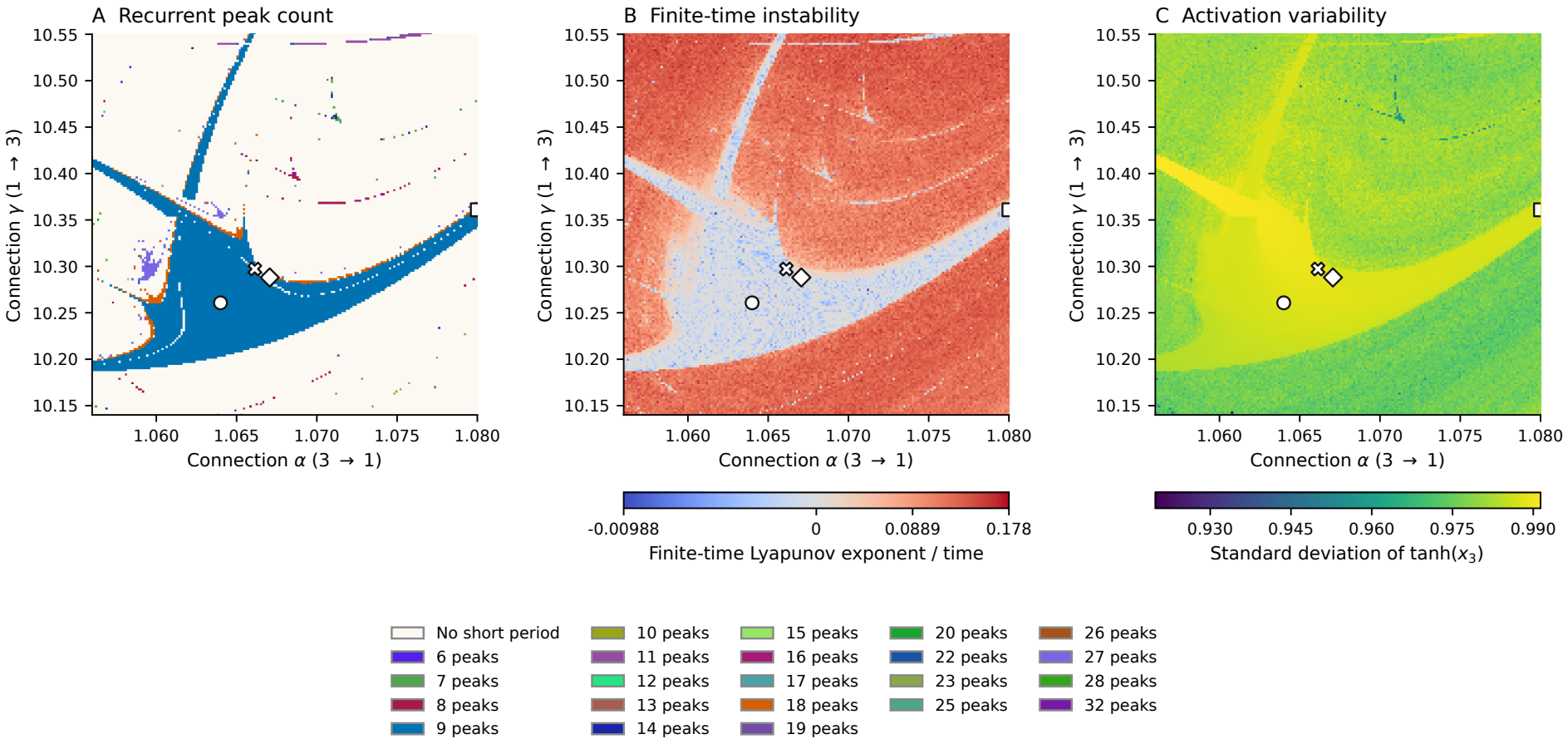


Figure 2: The same geometry in measured network quantities. The focused controls are identical in all three panels: (A) full-state peak recurrence, (B) finite-time Lyapunov estimate per unit time, and (C) standard deviation of neuron 3's actual activation $\tanh(x_3)$. The low-exponent window and activation-statistic structure align visibly with the periodic organization. Panel C uses its labeled narrow range: large color contrast does not imply an equally large absolute activation change. Its statistic summarizes variability and cannot represent the full temporal waveform. A near-zero finite-time exponent in a periodic autonomous flow is consistent with a neutral time direction; positive finite-horizon values alone do not certify asymptotic chaos. All sampled cells and original classifications are retained.

3 How the geometry appears in neural activity

3.1 An independently initialized transect

Figure 3 follows the same connection γ across measured potential maxima, complete-state recurrence, sensitivity, and all three activation variabilities. The recurring branches occupy settings with much smaller finite-time sensitivity than the surrounding irregular sequences. The activation standard deviations change more modestly: the strongest change is in temporal organization, which a single mean or variance cannot fully describe.

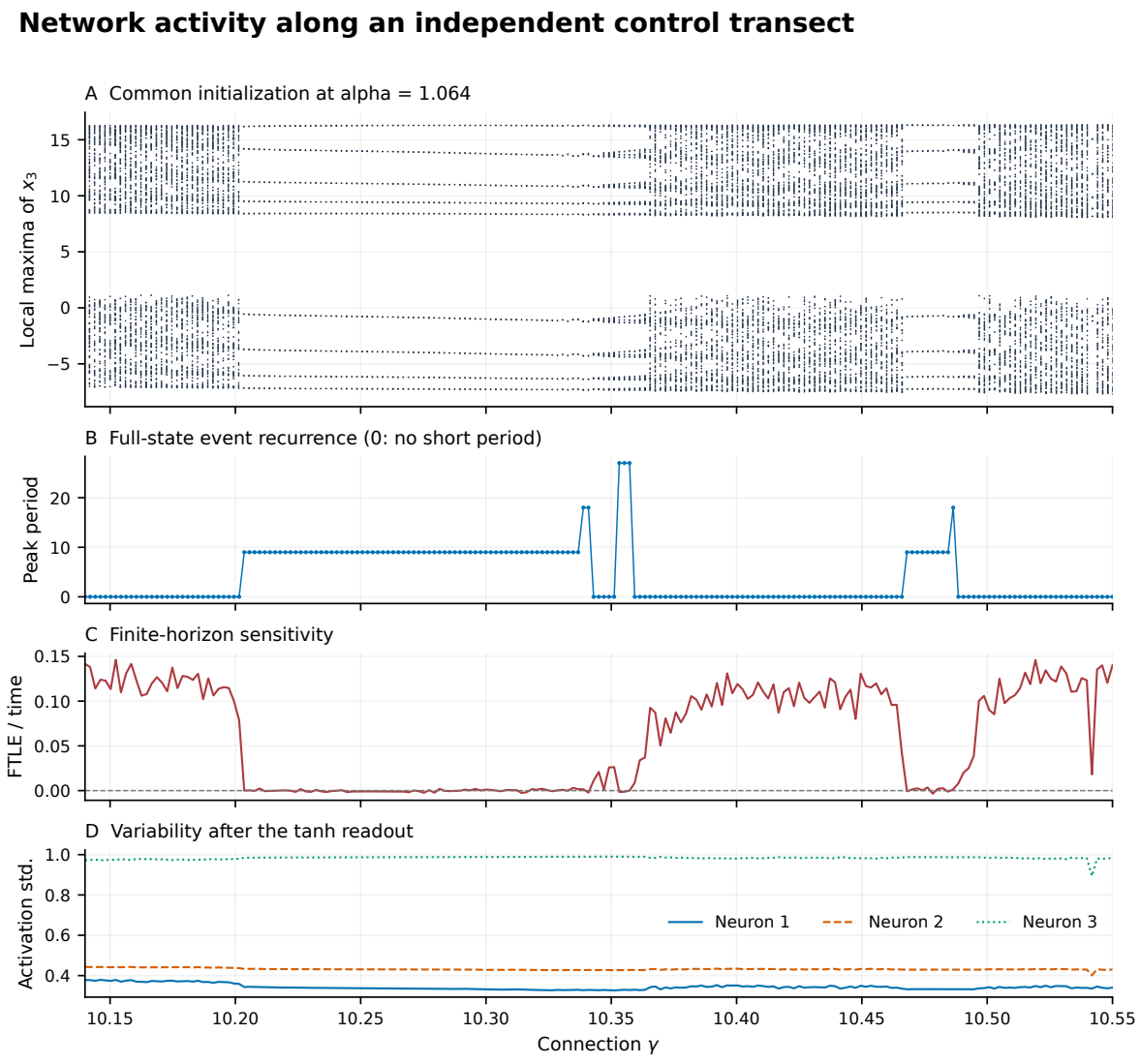


Figure 3: Common-initialization neural transect. At fixed $\alpha = 1.064$, all 201 points independently begin at $(0.1, 0.1, 0.1)$. (A) Every retained local maximum of x_3; (B) candidate peak period from complete-state event recurrence; (C) finite-time Lyapunov estimate; (D) activation standard deviations for the three neurons. A zero period retains the unresolved category. Lines connect measured settings and do not locate exact bifurcation boundaries. This transect uses twice the grid settling time and a smaller step, with no continuation from neighboring controls.

3.2 The full waveform and the effect of saturation

At the verified core, the complete state repeats every 37.63637538 time units, with nine maxima of x_3 in that cycle. The separately integrated behavior probe reports one-period state repetition RMS 8.15×10^{-10} from 2,248 dense-output pairs in its final half. Figure 4 displays every neuron and its output, so the parameter label is connected directly to network motion.

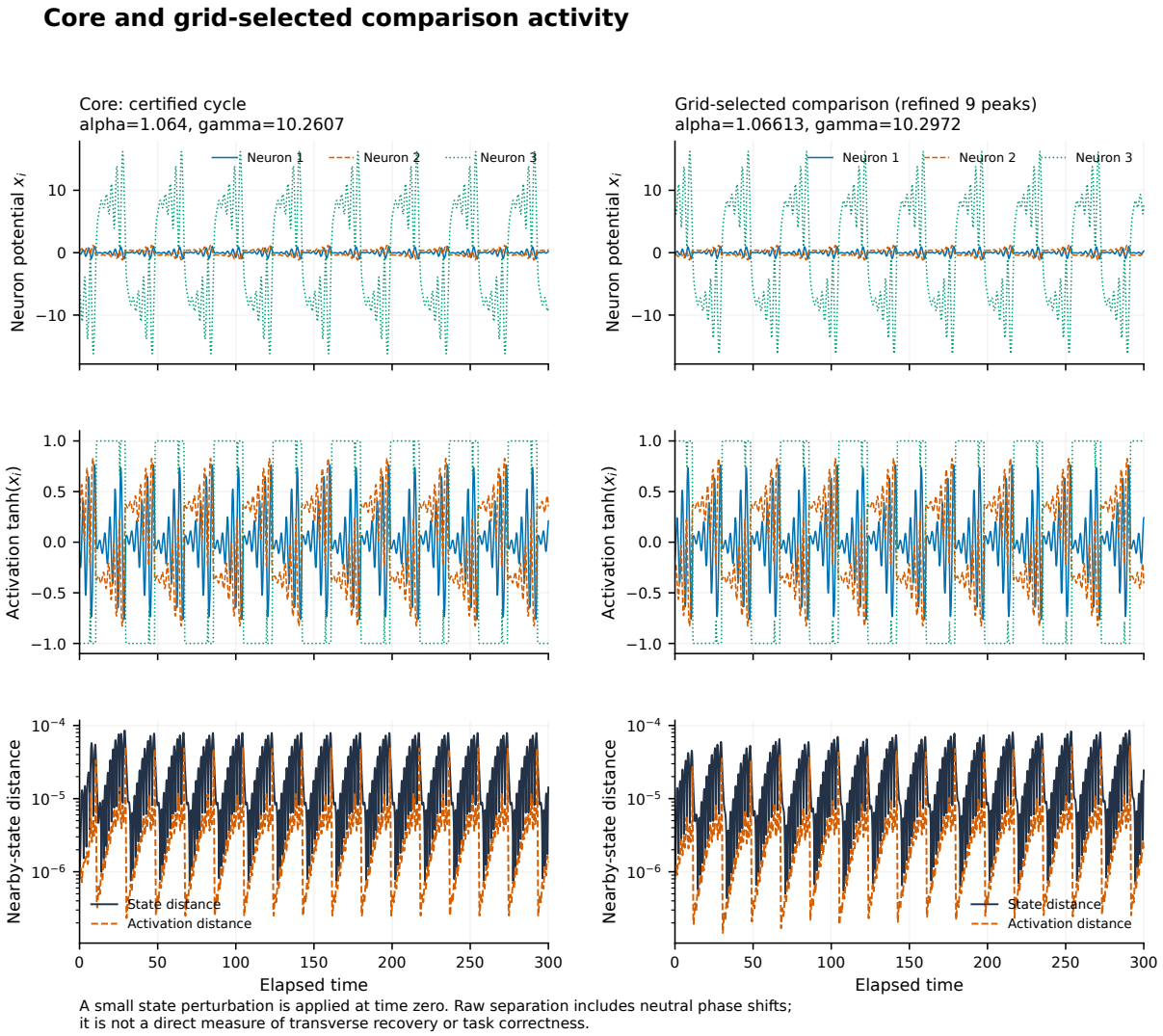


Figure 4: Neuron potentials, activations, and paired perturbations. The core is a numerically verified attracting cycle. The comparison was selected as an exterior by the original grid, but both refinements detect nine peaks; it is *not* a validated chaotic exterior. Columns retain those original roles. Rows show all three potentials, their tanh activations, and same-time state/activation separation after a $10^{-6}(1,1,1)/\sqrt{3}$ perturbation. Every stored time over 300 units is plotted. Persistent or temporarily growing same-time separation can contain a neutral phase shift and transient amplification; it is not a direct measurement of transverse recovery. The failed exterior interpretation is retained rather than replaced by a more favorable example.

The complete activation readout is componentwise injective for finite states: $\tanh(x(t+T)) = \tanh(x(t))$ exactly when $x(t+T) = x(t)$. It therefore preserves the full waveform's primitive period.

Saturation changes amplitude and numerical visibility, not this exact correspondence. At the core, neuron 3's potential spans approximately -16.20 to 16.20 in the recorded behavior window, while its activation approaches ± 1. Over the final 150–300 time units of the 300-unit probe, its population potential and activation standard deviations are respectively 8.664 and 0.9876; these have different units and are reported separately.

4 What survives numerical refinement

4.1 Three independently verified periodic orbits

Table 1 retains the coarse and refined peak counts at every selected coordinate. The core and outer-component point remain nine-peak candidates, and the doubled point remains an eighteen-peak candidate at both smaller steps. Their independently shot periods agree closely with the refined estimates. Full-state closure and the neutral-mode check distinguish these observations from repetition of a scalar loss or a projected trajectory.

Selected role	α	γ	Peaks: .02/.01/.005	Shot T	Closure
Core	1.064000	10.260722	9/9/9	37.636375	7.46×10^{-10}
Outer	1.080000	10.360944	9/9/9	36.974021	1.46×10^{-9}
Doubled	1.067067	10.288056	18/18/18	74.481699	9.69×10^{-11}
Exterior	1.066133	10.297167	0/9/9	—	—

Table 1: Refinement and complete-state closure. Coordinates are rounded for display; exact grid coordinates are retained in the manifest and selected evidence. Closure is $\|\phi_T(x_0) - x_0\| / \max(1, \|x_0\|)$. Zero is no detected short period. The exterior failed the frozen requirement that the original and both refined peak counts agree, so no shooting certificate was issued for it. It remains part of the result.

Orbit	Neutral flow multiplier	Transverse multiplier 1	Transverse multiplier 2
Core	1.000000015	0.06506010	6.15×10^{-11}
Outer	1.000000001	-0.46513914	-7.81×10^{-12}
Doubled	1.000000080	0.65612133	-3.45×10^{-19}

Table 2: All three numerical Floquet multipliers. The neutral direction is retained. Attraction refers to the two transverse directions. Values near zero are numerical approximations with ill-conditioned tiny magnitudes; their final digits and signs should not be interpreted as precisely resolved. Contraction factors are per complete cycle, whose duration differs across rows, not directly comparable rates per unit time.

The core's dominant transverse multiplier is 0.06506, while the outer point's is -0.46514. These describe different local perturbation responses even though both have nine peaks. They do not estimate basin volume, noise tolerance, or recovery from an arbitrary disturbance. The selected outer point reaches the right edge of the focus rectangle, so it cannot establish where that extension terminates. A nearby eighteen-peak orbit is consistent with doubled periodic organization, but this experiment does not continue an exact bifurcation branch connecting the two.

4.2 The exterior control exposes a finite-horizon error

The coarse-grid exterior has no detected short period and a finite-time exponent of 0.00558. Both refinements instead detect a nine-peak sequence. Its captured behavior therefore cannot support

an inside-versus-chaos claim. The change could reflect longer settling, numerical refinement, or their combination; the design changes both and does not isolate the cause. This failure is especially relevant near window boundaries, where a finite map can mistake a long transient for persistent irregular dynamics.

The perturbation traces also require the flow interpretation. At the core, a 10^{-6} perturbation reaches maximum same-time state separation 8.53×10^{-5} over 300 units, despite the attracting transverse multipliers. A neutral phase displacement and transient amplification can coexist with orbital attraction. The saved nearest-orbit distances include finite orbit-sampling error and are not exact phase projections. An inference of useful disturbance rejection requires a dedicated phase-aware measurement.

5 Independent controls show a large activity contrast

EXP-005 is a prospective follow-up chosen after the EXP-004 failure and frozen before its own outcomes. It fixes the earlier core, the two endpoints of the independent transect, and the failed near-edge comparison. Each point independently restarts at $(0.1, 0.1, 0.1)$ at both $\Delta t = 0.01$ and 0.005, settles for 10,000 time units, and is measured for 2,000. The finite-time exponent uses the final 1,900 units after tangent alignment. The two distant controls meet the frozen irregularity rule at both steps: no short recurrence and exponent above 0.005. This supports persistent irregularity over the tested horizons, not an asymptotic-chaos proof.

The core again has a verified nine-peak orbit: $T = 37.63637538$, relative closure 8.55×10^{-11}, and transverse spectral radius 0.06506. The failed near-edge point also receives an independent nine-peak certificate, with $T = 37.33149959$, closure 6.46×10^{-10}, and transverse spectral radius 0.85603. EXP-005 requires agreement between its two fresh runs; it does not require agreement with the original coarse grid. Its new certificate therefore does not erase the EXP-004 failure.

From each finest-step terminal state, the same 10^{-6} state perturbation is applied in direction $(1, 1, 1)/\sqrt{3}$. Figure 5 and Table 3 show the resulting 300-unit probes. The core's final activation separation is 1.53×10^{-6}, versus 1.41 and 2.13 at the lower and upper controls; their maximum state separations reach 29.67 and 31.87. These are actual changes in the network's outputs, with a contrast of about six orders of magnitude at the final checkpoint. They measure same-time repeatability for one declared direction and phase per setting, not population robustness, exact phase-aligned recovery, or useful task performance.

Control	γ	Peaks: .01/.005	FTLE: .01/.005	$\|\Delta x(300)\|_2$	$\|\Delta u(300)\|_2$
Core	10.260722	9/9	$-0.000410/-0.000411$	1.66×10^{-5}	1.53×10^{-6}
Lower	10.140000	0/0	0.138663/0.132212	9.850	1.408
Upper	10.550000	0/0	0.122586/0.135211	16.137	2.129
Failed near-edge	10.297167	9/9	0.001239/0.001239	3.24×10^{-5}	3.42×10^{-6}

Table 3: All four EXP-005 controls. $\alpha = 1.064$ except the retained near-edge coordinate, $\alpha = 1.0661333333$. FTLE is per time unit over 1,900 units; separation uses a distinct 300-unit paired probe. Zero means no detected short period.

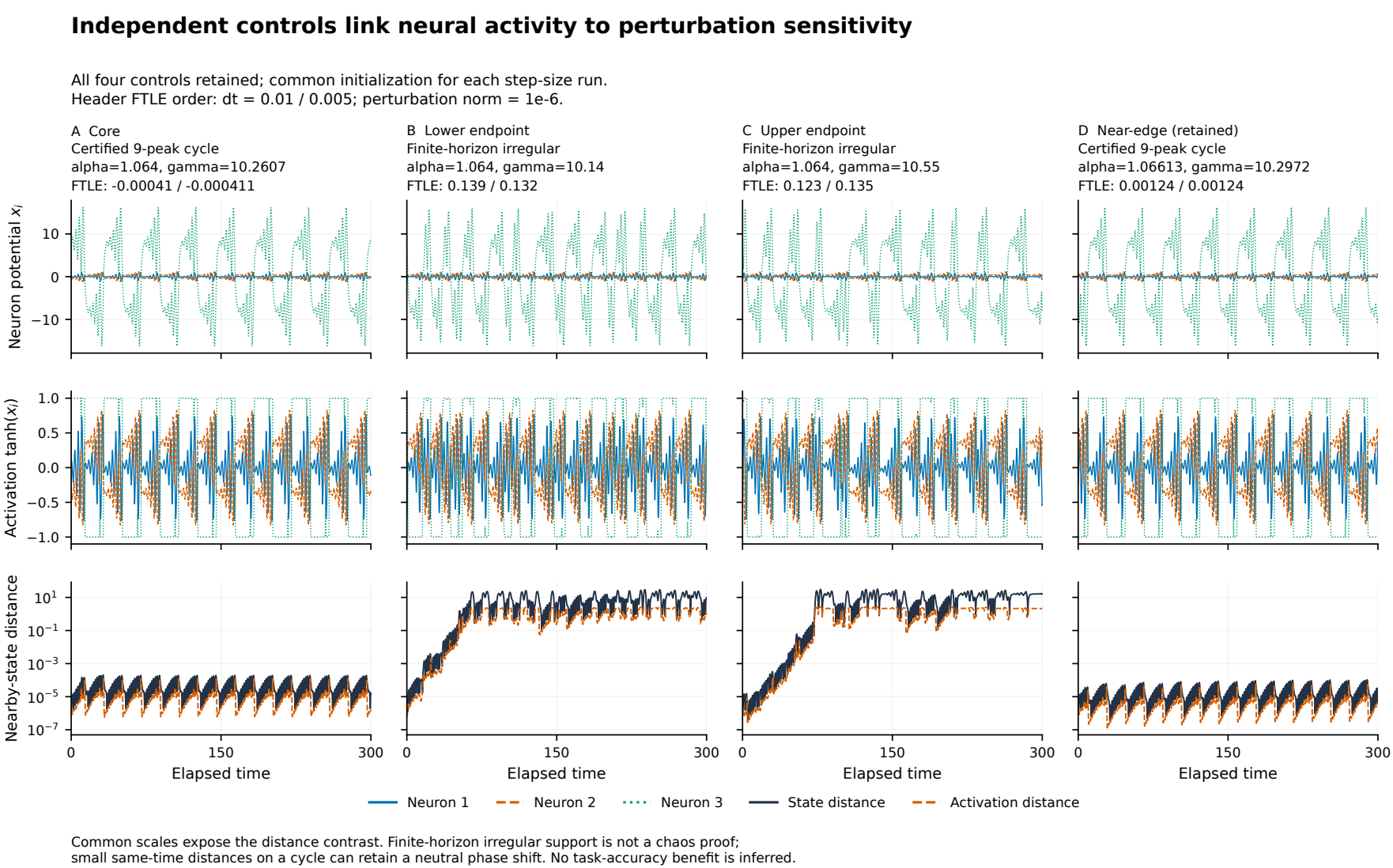


Figure 5: Independent long-settling controls: the parent network behaves differently. EXP-005 retains all four settings: core, lower and upper transect endpoints, and the failed near-edge comparison. Rows show the three neuron potentials, their actual tanh activations, and paired state/activation separations over 300 units. The two distant controls remain irregular by the prespecified finite-horizon rule at both integration steps. Both other settings have independent attracting-cycle certificates. The near-edge column records a new experiment at the failed coordinate; Figure 4 and Table 1 preserve its earlier outcome. All columns use the same perturbation norm and direction; initial phase is the independently obtained terminal state.

6 What the neural shrimp tells us, and what to test next

The shrimp selects a repeating activation waveform and accompanies a strong difference in measured sensitivity. Whether to enter or avoid it depends on the desired computation: a rhythm generator could use a periodic operating region, while a solver intended to settle to one answer could be impeded by cycling. No task endpoint is built into this network.

The clearest untested use of the *shape* is tolerance to changes in connection weights. A transverse Floquet multiplier describes a small change of *state at fixed weights*; window width describes where periodic organization persists as *weights change*. Moving a network state does not move its weights across the shrimp map. The next tests should therefore separate those two disturbances:

1. **Does the shape predict tolerance to weight drift?** Continue the verified branches and locate stability boundaries, retaining failures and unstable segments. At prospectively chosen body, extension, and boundary settings, perturb the two weights in matched directions and test multiple initial states and phases. Report doses both in raw units and a frozen dimensionless metric. Test whether boundary distance or window width predicts regime loss and waveform error beyond current period, amplitude, clean waveform quality, and Floquet multipliers on held-out settings.

2. **Can this improve rhythm control?** Freeze a reference activation waveform, allowable phase error, and separate evaluation disturbance streams. Compare a controller using geometry with ordinary parameter search and controllers given the same local cycle diagnostics, actions, observations, and total resources. Include random directions, a shuffled-boundary control, and zero-dose numerical checks. Measure time within waveform/phase tolerance, recovery, effort, and total cost including the map. Separately perturb states at fixed weights and measure phase drift and phase-aligned orbital recovery.

3. **Does the mechanism transfer?** Reproduce one small training or reasoning model before testing entry, exit, averaging, or stopping. Its full evolving state must include optimizer buffers or secondary latents as appropriate. Measure held-out predictions, correctness, settling time, and cost; recurrence alone does not establish useful computation.

The present map reproduces recognizable shrimp-shaped organization and verifies selected neural orbits; it does not certify every pixel or the exact boundary topology. Compared with the published 1000×1000 scans, $\Delta t = 0.01$, and settling time 10,000, our scan has lower resolution and shorter settling. Source initial states and row-reset conventions were unspecified; ours are explicit. The printed source transect equation appeared inconsistent with its displayed range, so we used the independently frozen vertical transect. These differences delimit reproduction rather than being hidden numerical substitutions.

Evidence, cost, and research status

The code, protocol, and machine plan were pushed at revision `b670086775bb` before EXP-004 outcomes. Its one completed attempt used four local CPU workers and 140.54 seconds of measured wall time. EXP-005 was frozen and pushed at revision `01209371d541` before its eight fresh runs; its completed attempt used one worker and 4.17 seconds. Recorded paid compute and paid review costs are zero. Raw row states, peak events, statistics, selected probes, refinements, and certificates are retained with source hashes. Figures retain plotted arrays and source/output receipts. The plotting tools render saved data without advancing the ODE. Both protocols are in `docs/`, and exact plans

are in `experiments/hopfield/`. Complete attempts are retained under `evidence/EXP-004/` and `evidence/EXP-005/`; `docs/next-steps.md` specifies the proposed utility tests. Private Git history supports internal chronology, not public preregistration.

Generative-AI tools assisted with design, implementation, numerical execution, audits, plotting, and drafting. The human selected the question; authorship and final release approval remain pending. No paid Pro review was requested or run.

A Earlier pilot history

EXP-001–003 remain unchanged and are not evidence for the neural shrimp reproduced here. The Hénon control verified nontrivial attracting cycles in 39 of 65 cells. The tiny tanh *training* scan found 0 verified cycles in 162 cells, with 107 unresolved and 55 divergent. In the 96-run policy pilot, the recurrence detector never intervened; each policy had 9 divergent cases among 24 pairs. The old plots, exact scalar toy, and receipts remain in the repository as exploratory history. EXP-004 supplies a separate, measured recurrent-network periodic window; it does not retroactively validate those training hypotheses.

References

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